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Source-Stratified Evaluation of Out-of-Distribution Detection in Fruit-Quality Classification Under Species and Treatment-Labelled Shifts

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Abstract

Automated fruit-quality classification using computer vision can achieve strong performance on familiar image distributions but may become unreliable under changes in data sources, fruit species, and post-harvest treatment conditions. This study presents a source-stratified empirical evaluation of a frozen MobileNetV2-based six-class fruit-quality classifier and multiple out-of-distribution (OOD) detection approaches. The evaluation uses 13,465 image records from three image repositories, organized into primary in-distribution, external known-class, unseen-species, treatment-labelled, and combined-shift cohorts.

Five established OOD scoring approaches—Maximum Softmax Probability (MSP), Energy, ODIN, Mahalanobis distance, and OpenMax—are compared with an operational Energy–MSP dual-gate cascade. Closed-set accuracy decreases from 96.92% on the primary in-distribution test partition to 68.36% on the pooled external known-class partition, reflecting a 28.56 percentage-point difference across evaluation sources. Within the FruitVision repository, Energy achieves an AUROC of 81.57% for the known-versus-unseen-species comparison, with an operational false acceptance rate of 13.18%, whereas the observed AUROC on the source-matched Sultana unseen-species comparison was 67.52%. In the evaluated FruitVision treatment-labelled cohort, discrimination between natural known-class images and treatment-labelled known-class images is limited, with Energy achieving 53.13% AUROC. Under fixed dual-gate operating thresholds calibrated on validation data, 42.83% of the treatment-labelled image records are accepted and assigned a Fresh class.

The findings demonstrate that post-hoc OOD detection performance varies substantially across distribution shifts and data sources. In the evaluated treatment cohort, tested RGB post-hoc scoring methods did not reliably separate images recorded under the dataset-reported treatment protocol from natural fruit. The study provides an image-level benchmark analysis with source-stratified comparisons, uncertainty estimates, duplicate-sensitivity analysis, and explicit limitations concerning dataset labels, specimen independence, and operational validation.

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Keywords: Fruit quality classification; Open-set recognition; Out-of-distribution detection; Domain shift; MobileNetV2; Energy-based detection; Image classification; Food quality assessment

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1. Introduction

1.1. Research Background and Motivation

Post-harvest agricultural loss represents a critical challenge for global food security, economic stability, and supply chain sustainability [1]. According to the Food and Agriculture Organization (FAO), approximately 14% of global food production is lost between harvest and retail distribution [1]. In regional agrarian economies, post-harvest deterioration across domestic transit corridors is particularly acute. For example, nationwide surveys in India conducted by the Indian Council of Agricultural Research (ICAR-CIPHET) indicate that post-harvest losses of commercial fruits consistently range between 6.70% and 15.88% (e.g., 10.39% in apple, 7.76% in banana, and 15.88% in citrus) [2]. These losses disproportionately penalize smallholder farmers and aggregation packhouses, where produce sorting and quality grading have historically relied upon manual, visual inspection. Manual grading is inherently subjective, labour-intensive, prone to operator fatigue, and incapable of scaling to high-throughput sorting facilities [3, 4].

In response to these operational bottlenecks, computer vision and deep learning have emerged as promising non-destructive methodologies for automated fruit freshness assessment, ripeness grading, and surface defect classification [3–6]. Convolutional neural networks (CNNs), particularly lightweight architectures such as MobileNetV2 [7], have become central to practical agricultural deployments. Depthwise separable convolutions and inverted residual bottlenecks allow MobileNetV2 to execute on resource-constrained mobile phones and edge sorting devices with parameter budgets under 4 million weights [5, 7]. In controlled academic evaluations on curated benchmarks, reported top-1 accuracies often exceed 95% on specific multi-class produce datasets [4–6].

1.2. The Closed-Set Assumption and Its Vulnerabilities in Agricultural Vision

Despite high benchmark accuracy, practical deployment of deep learning models in real-world agricultural environments encounters fundamental reliability barriers. Standard supervised image classifiers operate under a strict closed-set assumption: the training, validation, and operational deployment data are presumed to be independent and identically distributed (i.i.d.) over a fixed, finite set of semantic classes, $\mathcal{Y} = \{1, 2, \ldots, K\}$ [8, 9]. In practical agricultural workflows, however, this assumption rarely holds. Vision models deployed at packhouses, retail terminals, or farm collection centres confront three distinct, compounding categories of distribution shift:

1. Covariate and Domain Shift: Images collected across external facilities exhibit variations in background illumination, sensor optics, specular glare, ambient shadows, and post-harvest handling marks that differ from training sets [3, 5, 10, 11]. Furthermore, natural biological diversity in produce appearance, surface texture, and pigmentation across agricultural batches can induce domain shifts that affect learned feature representations [11, 12].
2. Semantic Shift (Unseen Species): In automated sorting lines, bins, or field crates, models trained exclusively on a small palette of target crops (e.g., apple, banana, and orange) frequently encounter non-target produce species (e.g., mango, guava, jujube, pomegranate, or lime). Under standard closed-set inference, deep networks lack an explicit rejection mechanism; they are forced to assign an unseen species to one of the K known classes, often with uncalibrated, erroneously high confidence [8, 9].
3. Treatment-Labelled Condition Shift: In certain agricultural supply-chain and research contexts, produce may be subjected to chemical preservative treatments or immersion protocols intended to retard visible senescence. In curated benchmarks such as FruitVision [12], images are documented under a specific dataset-reported post-harvest treatment protocol involving immersion in a 10% formalin and 90% water

solution for 30 minutes, followed by drying and photography 48 hours later. Whether visual models trained on natural produce can reliably distinguish images recorded under such post-harvest treatment protocols remains an empirical question.

Under standard softmax classification, neural networks are notoriously overconfident [8]. Logit outputs normalized via the softmax function produce probability distributions that often peak sharply near 1.0 even when the input image is heavily corrupted, semantically anomalous, or completely out-of-distribution [8, 9]. In automated quality sorting, an overconfident false-positive classification—such as misclassifying an unseen non-target species or an anomalous produce lot as a fresh known-fruit class—presents substantial commercial and quality-assurance liabilities.

1.3. Open-Set Recognition and Out-of-Distribution Detection Foundations

To address closed-set failure modes, computer vision research has developed two closely related paradigms: Open Set Recognition (OSR) [13, 14] and Out-of-Distribution (OOD) Detection [9, 15].

The concept of open-set recognition was formalized by Scheirer et al. [13], who defined the objective as simultaneously classifying known training classes while bounding the open space risk—the risk of incorrectly labeling an unknown, novel class sample as any known class. Bendale and Boult [14] extended this concept to deep neural networks through OpenMax, which estimates the probability of an input belonging to an unknown class by fitting Weibull distributions to activation vectors in the penultimate layer.

Simultaneously, the broader machine learning community formalized generalized out-of-distribution detection [9, 15]. Hendrycks and Gimpel [9] established the Maximum Softmax Probability (MSP) baseline, showing that correctly classified in-distribution samples typically exhibit higher maximum posterior probabilities than anomalous inputs. Liang et al. [16] developed ODIN, which amplifies the separation between in-distribution and out-of-distribution samples using temperature-scaled softmax outputs and input-space gradient perturbations. Lee et al. [17] introduced a feature-space Mahalanobis distance metric that models class-conditional Gaussian distributions across intermediate hidden representations. More recently, Liu et al. [18] proposed Energy-based OOD detection, utilizing the log-sum-exp of the classifier logits to circumvent the probability saturation that characterizes standard softmax normalization [15, 18]. Additionally, Vaze et al. [19] demonstrated that closed-set classifier capacity and representation quality strongly dictate open-set discrimination performance.

In operational deployment, OOD detectors are typically implemented either as continuous ranking functions or as discrete rejection gates governed by a pre-calibrated threshold τ. When an input sample produces an anomaly score that violates the threshold, the system abstains from classification, triggering manual inspection or secondary non-optical assay.

1.4. Prior Agricultural OSR/OOD Studies and Specific Research Gaps

Open-set recognition and anomaly detection have recently attracted growing attention in agricultural vision. Several recent investigations have examined related challenges:

- Sattar et al. [20] developed SensorNet, combining computer vision, deep learning, and chemical sensors to identify chemical adulteration in fruit specimens.
- Bijoy et al. [12] introduced FruitVision, an image dataset containing fresh, rotten, and formalin-mixed fruit images.
- Wang et al. [21] proposed lightweight attention networks with unsupervised domain adaptation for cross-domain fruit classification.
- Pehlivan and Kara [22] introduced a lightweight one-shot open-set metric learning framework for food recognition in smart appliances.
- Jansi and Kavitha [23] developed PEOS-Net, a prototype-energy open-set network for robust crop leaf disease recognition beyond the closed-world assumption.

- Yan et al. [24] formulated PAG-MSBNet, using prototype-aware gated multi-scale branch networks for open-set crop disease recognition.

While these studies highlight the importance of non-standard conditions in agricultural classification, existing research has not systematically evaluated how established post-hoc OOD scoring functions behave when a frozen classifier faces multi-source external distributions. Specifically, four key empirical gaps remain:

1. Gap 1: Limited Evaluation Beyond Dataset-Specific Closed-Set Splits: Existing fruit-freshness studies predominantly report supervised classification performance on random train/test partitions within a single, curated dataset [4–6]. Such closed-set protocols evaluate neither model behavior on external image repositories nor discrimination against out-of-distribution instances. Furthermore, recent evaluations of agricultural image collections indicate that random partitioning within single repositories can allow near-duplicate images across splits, yielding optimistic closed-set performance estimates that do not reflect external test conditions [10, 11].
2. Gap 2: Cross-Repository Confounding in OOD Benchmarks: In open-set evaluations where out-of-distribution cohorts are drawn from a different repository than the known-class training data, differences in camera hardware, ambient illumination, and background context present a substantial methodological risk. Under such unstratified designs, repository-level acquisition artifacts can be confounded with true semantic divergence between fruit species [15]. Evaluating known and unknown categories collected under shared repository conditions (source matching) reduces repository-level confounding, though potential within-repository capture variations may remain.
3. Gap 3: Empirical Behavior Under Treatment-Labelled Shifts: While multimodal sensing has explored chemical adulteration detection combining optical imaging with specialized gas sensors [20], the behavior of post-hoc OOD scoring methods on a standard frozen RGB classifier facing dataset-reported treatment conditions [12] without auxiliary chemical sensors warrants systematic empirical investigation.
4. Gap 4: Distinguishing Retrospective Ranking from Operational Threshold Calibration: Many OOD detection studies primarily report retrospective metrics: AUROC assesses ranking discrimination across all decision thresholds without selecting an operating point, whereas FPR@95 evaluates the false positive rate at a retrospective threshold chosen post-hoc to achieve 95% true positive rate on the test cohort [9, 15, 18]. While suitable for relative algorithm comparison, retrospective metrics do not reflect operational deployment, where a discrete acceptance threshold must be pre-calibrated on validation data and held fixed during external evaluation. Measuring operational Unknown False Acceptance Rates (UFAR) alongside Known-Class Rejection Rates (KRR) under fixed validation thresholds provides complementary, deployment-context measurements of operational trade-offs.

1.5. Research Objectives and Formal Research Questions

This investigation addresses these gaps by conducting an extensive, source-stratified empirical evaluation of out-of-distribution detection in fruit image classification. The study evaluates a frozen MobileNetV2 six-class visual classifier across $N = 13,465$ image records curated from three distinct repositories under the Stage 4.0.3 external evaluation protocol.

The paper specifically investigates three core research questions:

- Research Question 1 (RQ1 — External Generalization): How does the closed-set classification performance (top-1 accuracy, macro-F1, and balanced accuracy) of a lightweight MobileNetV2 model change when evaluated on external repositories containing the same known fruit classes?
- Research Question 2 (RQ2 — OOD Discrimination): How effectively do established OOD scoring mechanisms (MSP, Energy, ODIN, Mahalanobis distance, OpenMax) discriminate between in-distribution produce and distribution shifts when evaluated in source-matched cohorts examining (a) unseen-species semantic shift, (b) treatment-labelled condition shift, and (c) combined species-and-treatment shifts?

- Research Question 3 (RQ3 — Operational Reliability): How do fixed validation-calibrated rejection thresholds (τ_{95}) affect out-of-distribution false acceptance rates (UFAR) and known-class rejection rates (KRR) under external domain and condition shifts? (Where known-class retention is defined as $1 - \text{KRR}$.)

1.6. Evidence-Supported Research Contributions

The primary contribution of this work is an empirical source-stratified evaluation, rigorous uncertainty audit, and diagnostic failure analysis. The study does not propose a novel standalone OOD algorithm, nor does it claim real-world agricultural supply chain deployment. The four specific, evidence-backed contributions of this study are:

1. Audited Source-Stratified Benchmark Census ($N = 13,465$): We establish an audited image-level evaluation benchmark comprising 13,465 image records structured across five distinct partitions: Partition A (Primary ID Test, $N = 227$), Partition B (External Known-Class Shift, $N = 5,350$), Partition C (Treatment-Labelled Condition Shift, $N = 1,917$), Partition D1 (Unseen-Species Shift, $N = 4,776$), and Partition D2 (Combined Shift, $N = 1,195$). We incorporate exact-duplicate handling and perceptual-hash (pHash) sensitivity analysis across Hamming distance radii $r \in \{0, 4, 8\}$.
2. Quantification of Closed-Set Performance Differences Across Partitions: We evaluate the frozen MobileNetV2 classifier across partitions, observing a 28.56 percentage-point difference in top-1 accuracy between the primary in-distribution test partition (Partition A: 96.92% [94.71%, 98.68%]) and the pooled external known-class partition (Partition B: 68.36% [67.20%, 69.40%]), with balanced accuracy decreasing from 97.09% to 68.66%. Through source stratification, we show that performance varies across external repositories (FruitVision B: 69.14% accuracy, 69.56% balanced accuracy; Sultana B: 65.63% accuracy, 65.55% balanced accuracy), with closed-set errors heavily concentrated in the fresh apple class (where top-1 accuracy drops to 32.12% across pooled external images due to confusion with rotten apple and rotten orange).
3. Source-Matched Evaluation Across Semantic and Treatment Cohorts: By benchmarking within shared repositories, source matching reduces cross-repository confounding while potential within-repository capture variation remains. For the Within-FruitVision known-versus-unseen-species comparison (Cohort A), Energy-based detection achieves an AUROC of 81.57% [80.81%, 82.38%] with an operational UFAR of 13.18% [12.07%, 14.34%] ($366/2,776$ image records at $\tau_{\text{Energy}} = 1.054579$), whereas discrimination was lower on the Within-Sultana comparison (Cohort D: Energy AUROC of 67.52% [65.67%, 69.25%], operational UFAR of 34.15% [32.35%, 35.90%], $683/2,000$ image records).
4. Empirical Evaluation of Treatment-Labelled Images and Operational Trade-Offs: In the evaluated FruitVision treatment-labelled cohort (Cohort B), the frozen classifier and tested post-hoc OOD scoring methods did not reliably distinguish images labelled under the dataset-reported treatment condition from the natural known-class cohort, with Energy achieving an AUROC of 53.13% [52.16%, 54.09%]. Under locked validation operating thresholds (τ_{95}), the Dual-Gate cascade yields an operational UFAR-Fresh of 42.83% [41.26%, 44.34%] ($821/1,917$ image records), accepting and assigning a Fresh class to these images. Furthermore, for the Dual-Gate cascade under τ_{95}, an operational UFAR of 16.07% [14.77%, 17.40%] ($446/2,776$ image records) on FruitVision unseen species incurs an external known-class rejection rate (KRR) of 41.98% [40.49%, 43.49%] ($1,745/4,157$ image records, corresponding to a retention rate of $1 - \text{KRR} \approx 58.02\%$) on FruitVision Known, and 42.43% [41.10%, 43.76%] ($2,270/5,350$ image records) across pooled external known produce (Partition B).

2. Related Work

This section reviews the scientific literature foundational to the present study across four key domains: (1) computer vision methodologies for fruit quality assessment and freshness classification; (2) cross-domain generalization and domain adaptation in agricultural vision systems; (3) established open-set recognition

and out-of-distribution (OOD) detection algorithms; and (4) recent applications of open-set and anomaly detection frameworks to food and agricultural imagery. The section concludes with a structured comparison matrix and an explicit definition of the research gap addressed by this study.

2.1. Computer Vision for Fruit Quality Classification

Automated post-harvest quality assessment has progressed significantly over the past two decades, transitioning from classical hand-crafted optical feature extraction to deep convolutional neural networks (CNNs) [3]. Early automated sorting systems relied predominantly on engineered representations of colour, morphological geometry, and surface texture [3]. Colour histograms computed across RGB, HSV, and CIE $L^*a^*b^*$ spaces served as primary indicators of commercial maturity, chlorophyll degradation, and surface pigmentation changes. Morphological descriptors—including planar area, eccentricity, circularity, and perimeter irregularity—were leveraged to identify mechanical malformations and physical grading deviations. Texture descriptors, such as Gray-Level Co-occurrence Matrices (GLCM) and local binary patterns (LBP), provided statistical representations of surface coarseness, fungal lesions, and skin shrivelling. However, these classical pipelines exhibited acute vulnerability to illumination variations, background reflections, camera sensor discrepancies, and natural biological phenotypic diversity within identical cultivars [3].

The advent of deep convolutional architectures enabled end-to-end hierarchical representation learning, capturing subtle non-linear manifestations of produce senescence, mechanical bruising, and microbial decay without manual feature engineering [3, 4]. Multi-task convolutional networks have demonstrated the ability to concurrently estimate fruit categories, ripeness stages, and defect severity levels from unified photographic inputs [4]. Specialized deep architectures have achieved high classification accuracy on curated benchmarks for distinguishing fresh from rotten produce across common commercial species [5, 6].

In post-harvest logistics, packing facilities, and farm-gate sorting stations, computational efficiency and hardware constraints represent primary deployment criteria. High-throughput sorting conveyors and handheld mobile grading devices cannot accommodate computationally intensive, high-parameter vision backbones. Consequently, lightweight architectures have become the standard for embedded agricultural vision [5]. In particular, the MobileNetV2 architecture [7] has been widely adopted due to its use of depthwise separable convolutions, inverted residual blocks, and linear bottleneck layers. With approximately 2.26×10^6 trainable parameters (at standard width multiplier $\alpha = 1.0$), MobileNetV2 dramatically curtails floating-point operations while preserving high discriminative capacity. Deep transfer learning—wherein backbones pre-trained on large-scale natural image datasets (such as ImageNet) are fine-tuned on target agricultural domains—has further enabled robust feature extraction even when localized training sets are constrained in sample size [5, 6].

Despite high reported accuracies under closed-set evaluation protocols, photographic image acquisition in practical agricultural environments introduces severe distributional challenges. Ambient lighting fluctuations, direct sunlight glint, variable artificial warehouse illumination, diverse photographic backgrounds, and camera sensor colour-balance differences introduce pervasive domain shift [10, 11]. The FruitQ repository introduced by Abayomi-Alli et al. [10] provided a structured multi-fruit dataset comprising 11 fruit varieties categorized into three freshness states (Fresh, Mild, and Rotten), evaluating five baseline deep learning architectures (ShuffleNet, SqueezeNet, EfficientNet, ResNet18, and MobileNetV2) under closed-set protocols.

The empirical evaluation conducted in the present study examines photographic produce records across multiple repositories, drawing upon two primary external dataset publications:

1. Sultana et al. (2022) [11]: Introduced a photographic dataset for recognizing fresh and rotten fruit specimens (Data in Brief, 44:108552). The repository encompasses 16 fresh and rotten categories representing eight fruit types: grape, guava, jujube, pomegranate, strawberry, apple, banana, and orange. Images were collected from retail fruit shops and authentic agricultural field environments under natural ambient lighting, with collection and condition labelling conducted with assistance from an agricultural domain specialist.

2. Bijoy et al. (2025) [12]: Introduced FruitVision, a benchmark dataset for fresh, rotten, and formalin-mixed fruit detection (Data in Brief, 61:111752). The dataset comprises five fruit species (apple, banana, mango, orange, and grape) evaluated across three dataset-reported visual conditions: fresh, rotten, and a treatment-labelled condition (immersion in a 10% formalin and 90% water solution for 30 minutes, followed by 48-hour drying and photography, as documented by the dataset authors). The present study evaluates these photographic records under their dataset-reported labels and does not conduct chemical assays or independently verify formalin residue. Image collection was conducted using three distinct mobile smartphone cameras across varied indoor and outdoor lighting environments. The published raw collection comprises 10,154 original photographs, with geometric and photometric augmentations expanding the published archive to 81,232 images.

Evaluation Census Clarification: We emphasize that the published raw and augmented archive sizes reported by dataset authors (e.g., the 10,154 raw or 81,232 augmented images in FruitVision) represent upstream repository totals. These figures must not be confused with the audited benchmark cohort evaluated in the present study. As detailed in Section 3, our evaluation utilizes an audited local census of exactly 13,465 evaluated image records (Primary Kaggle: 227; FruitVision: 10,045; Sultana: 3,193) across five dedicated partitions. Every evaluated image record is uniquely tracked through cryptographic hashes, exact duplicates are systematically audited, and perceptual near-duplicates are evaluated via sensitivity analysis across Hamming distance thresholds. We explicitly note that image records cannot be assumed to represent independent biological specimens.

2.2. Cross-Domain and External Generalization in Fruit Classification

Supervised machine learning algorithms are traditionally grounded in the closed-set, independent and identically distributed (i.i.d.) assumption, wherein training and test samples are drawn from an identical underlying probability distribution: $P_{\text{train}}(X, Y) = P_{\text{test}}(X, Y)$ [15]. In practical agricultural deployments, however, this assumption rarely holds due to diverse distributional shifts. To analyze these phenomena rigorously, we maintain precise distinctions among shift mechanisms [15]:

- Covariate Shift: The marginal input feature distribution varies between source and target domains ($P_s(X) \neq P_t(X)$) while the posterior class mapping remains invariant ($P_s(Y \mid X) = P_t(Y \mid X)$). In agricultural imaging, covariate shift arises from variations in camera sensor optics, ambient illumination, background surfaces, capture geometry, and regional cultivar appearance differences.
- Label Shift: The prior class distribution shifts between domains ($P_s(Y) \neq P_t(Y)$) while the class-conditional input densities $P(X \mid Y)$ remain identical.
- Concept Shift: The conditional relationship between visual features and semantic class labels changes between environments ($P_s(Y \mid X) \neq P_t(Y \mid X)$), such that identical visual appearances correspond to different quality states.
- Open-Set Semantic Shift: The target domain contains semantic classes whose support lies outside the source taxonomy: $\text{supp}(P_t(Y)) \nsubseteq \text{supp}(P_s(Y))$.

In this taxonomy, the FruitVision treatment-labelled cohort represents images of known fruit species carrying a dataset-reported formalin-mixed condition. We do not classify this condition as concept shift or a new semantic class, nor do we conflate post-harvest treatment-labelled shifts with label-space expansion. Rather, it represents an empirical condition shift wherein visual manifestations of post-harvest immersion are assessed against a classifier trained on natural produce, without implying independent chemical residue detection.

In agricultural image analysis, it is also essential to distinguish between domain adaptation and cross-domain evaluation without adaptation:

- Domain Adaptation: Methods in this paradigm actively adjust model parameters, feature representations, or latent feature alignments between a known source domain $\mathcal{D}_s$ and a target domain $\mathcal{D}_t$ during training or test-time calibration. Unsupervised domain adaptation (UDA) utilizes unlabelled target-

domain images to minimize divergence measures (such as Maximum Mean Discrepancy or Correlation Alignment) or employs adversarial domain discriminators to project source and target features into a domain-invariant latent space.

- Cross-Domain Evaluation Without Adaptation: Cross-domain evaluation without adaptation provides a diagnostic assessment of model performance on selected external repositories. It does not, by itself, establish performance under physical field deployment. Instead, this protocol evaluates the direct, unadapted generalization of a pre-trained classifier when exposed to external photographic datasets without access to target-domain training data, fine-tuning steps, or parameter updates, benchmarking baseline failure modes when adaptation is unavailable.

A notable contribution in agricultural domain adaptation is the study by Wang et al. (2023) [21], titled "A cross-domain fruit classification method based on lightweight attention networks and unsupervised domain adaptation" (Complex & Intelligent Systems, 9(4):4227–4247). Wang et al. proposed HAM-MobileNet, integrating a hybrid spatial-channel attention module into MobileNetV3, coupled with an unsupervised domain adaptation framework that minimizes a domain discrepancy loss combining Local Maximum Mean Discrepancy (LMMD) and Correlation Alignment (CORAL). Their framework was evaluated across two fruit classification benchmarks: (1) a grape classification dataset containing seven grape varieties across four domains (controlled environment, illumination variation, background variation, and internet images); and (2) a general fruit classification dataset containing 11 fruit types across two domains. Their findings demonstrated that active domain alignment significantly improves target-domain classification performance relative to unadapted baselines.

While Wang et al. [21] provide valuable insights into active domain alignment, their methodology differs fundamentally from the present study. Wang et al. focus on model training and domain adaptation to maximize transfer performance. In contrast, the present investigation evaluates the diagnostic failure modes of a common frozen MobileNetV2 classifier without adapting its weights to the target repositories. Rather than modifying model parameters, our study investigates: (1) the magnitude of external known-class accuracy differences across distinct repository sources; and (2) whether post-hoc out-of-distribution detection methods can reliably identify and reject out-of-domain samples before misclassification occurs. Furthermore, as the underlying datasets, species definitions, backbones, and evaluation protocols differ, reported numerical accuracy values cannot be compared directly across incompatible study protocols.

2.3. Open-Set Recognition and Out-of-Distribution Detection

When deployed in unconstrained real-world environments, computer vision models inevitably encounter visual inputs that deviate from their training distributions. The inability of standard closed-set classifiers to handle such inputs motivated the formalization of Open-Set Recognition (OSR) and generalized Out-of-Distribution (OOD) detection [13–15].

Scheirer et al. (2013) [13] formalized open-set recognition as balancing empirical risk minimization on known training classes against open-space risk—the risk of erroneously assigning unknown inputs to known semantic classes. Bendale and Boult (2016) [14] extended this principle to deep neural networks, demonstrating that deep classifiers suffer from intrinsic overconfidence: deep networks map inputs into linear feature spaces capped by a softmax activation function $\sigma(z)_k = \exp(z_k)/\sum_j \exp(z_j)$, which forces posterior probabilities to sum to unity. Consequently, an unfamiliar, anomalous, or unseen-class input can produce near-maximal softmax confidence if its projection aligns even slightly closer to one class weight vector than others, even when the input resides far from all known training clusters in latent space [8, 9, 19].

Generalized out-of-distribution detection expands open-set recognition beyond semantic class novelty, addressing covariate domain shifts, environmental background variations, and corrupted sensor inputs [15]. In this study, we systematically evaluate five established post-hoc OOD scoring algorithms, applied to a common frozen MobileNetV2 classifier without architectural alteration:

1. Maximum Softmax Probability (MSP) [9]: Formulated by Hendrycks and Gimpel (2017) as an established confidence baseline, MSP utilizes the maximum predicted posterior probability as the confidence

score:

$$S_{\text{MSP}}(x) = \max_{k\in\{1,\dots,K\}} \frac{\exp(z_k(x))}{\sum_{j=1}^{K}\exp(z_j(x))} \tag{1}$$

where $z_k(x)$ represents the logit for class k. While computationally straightforward, MSP represents maximum softmax confidence scoring rather than calibrated probability estimation, and it frequently fails under severe distribution shift due to softmax probability saturation on out-of-distribution inputs.

2. OpenMax [14]: Introduced by Bendale and Boult (2016), OpenMax replaces the standard softmax layer by modeling the distribution of penultimate-layer activation vectors for each known class using Extreme Value Theory (EVT). Class-conditional mean activation vectors (MAVs) are computed on training data, and Weibull distributions are fitted to the largest distances between training samples and their respective MAVs. For an operational input x, OpenMax computes class-specific rejection probabilities, modifies activation magnitudes, and redistributes activation mass to an explicit "unknown" class ($P_0(x)$). An input is accepted only if $1 - P_0(x)$ exceeds a pre-calibrated threshold.
3. ODIN (Out-of-Distribution detector for Neural networks) [16]: Liang et al. (2018) demonstrated that OOD detection can be significantly enhanced without retraining through two post-hoc operations: temperature scaling ($T > 1$) and input-space gradient perturbation. Logits are scaled by temperature T, smoothing the probability distribution, while the input image x is perturbed via gradient ascent:

$$\tilde{x} = x - \epsilon\,\text{sign}\left(-\nabla_x \log \frac{\exp(z_{\hat{y}}(x)/T)}{\sum_{j=1}^{K}\exp(z_j(x)/T)}\right) \tag{2}$$

where $\hat{y} = \arg\max_k z_k(x)$. This directional perturbation increases the maximum softmax probability of in-distribution instances substantially more than that of out-of-distribution instances, amplifying the separation margin.

4. Mahalanobis Feature Distance [17]: Lee et al. (2018) proposed modeling penultimate feature representations $f(x)$ as a mixture of class-conditional multivariate Gaussians with class centroids μ_k and a tied covariance matrix $\mathbf{\Sigma} = \frac{1}{N}\sum_{k=1}^{K}\sum_{i:y_i=k}(f(x_i)-\mu_k)(f(x_i)-\mu_k)^\top$. The anomaly score is defined as the negative minimum Mahalanobis distance to the nearest class centroid:

$$S_{\text{Maha}}(x) = -\min_{k\in\{1,\dots,K\}} (f(x)-\mu_k)^\top \mathbf{\Sigma}^{-1}(f(x)-\mu_k) \tag{3}$$

This distance quantifies the statistical deviation of the latent feature representation under the fitted class-conditional Gaussian feature model.

5. Energy-Based Out-of-Distribution Detection [18]: Liu et al. (2020) demonstrated that mapping logits directly to Helmholtz free energy provides a theoretically grounded OOD scoring function that avoids the normalisation artifacts of softmax probabilities. The negative free energy is defined as:

$$S_{\text{Energy}}(x) = -E(x;f) = T\cdot\log\sum_{k=1}^{K}\exp\left(\frac{z_k(x)}{T}\right) \tag{4}$$

The energy score is derived from the log-sum-exp of the classifier logits and is motivated by an energy-based interpretation of the classifier [18]. In this study, it is used as an OOD ranking score, not as a calibrated estimate of the true input marginal density.

Methodological Positioning: These five algorithms represent distinct mathematical principles: maximum softmax confidence scoring (MSP), extreme value activation modeling (OpenMax), temperature and input gradient perturbation (ODIN), multivariate Gaussian feature distance (Mahalanobis), and unnormalized logit thermodynamics (Energy). None of these algorithms was specifically engineered for agricultural imagery; they were designed and benchmarked on general vision tasks (CIFAR, SVHN, ImageNet) [15]. In this study, we apply these established post-hoc scoring functions to a common frozen MobileNetV2 produce classifier across source-stratified fruit cohorts. Furthermore, the operational Energy–MSP Dual-Gate cascade investigated in our operational analysis is an empirical decision policy combining Energy and MSP at validation order-statistic operating thresholds (τ_{95}), and does not represent a new theoretical OOD scoring algorithm.

2.4. Open-Set Recognition in Food and Agricultural Applications

Recognizing the vulnerability of closed-set classifiers in unconstrained operational settings, recent studies have begun exploring open-set recognition and anomaly detection within agricultural and food-processing domains.

Sattar et al. (2024) — SensorNet [20]: Titled “A comprehensive approach to detecting chemical adulteration in fruits using computer vision, deep learning, and chemical sensors” (Intelligent Systems with Applications, 23:200402), this study investigated the detection of chemical adulteration (specifically formaldehyde immersion) in fruits. Sattar et al. formulated SensorNet, a multimodal framework integrating captured fruit imagery with chemical-sensor measurements from an electronic sensor array sensitive to formaldehyde gas. An essential methodological distinction exists between SensorNet and the present investigation: SensorNet is a multimodal sensor-fusion system that integrates optical images with direct chemical-sensor measurements. In contrast, our study evaluates image-level classification on photographic records carrying dataset-reported chemical treatment labels, with no independent chemical measurements conducted in the present benchmark. Because the sensing modalities and target outputs differ fundamentally, numerical classification accuracy in SensorNet cannot be directly compared to optical OOD AUROC or operational false acceptance rates.

Pehlivan and Kara (2026) [22]: Titled “A Lightweight One-Shot Open-Set Metric Learning Framework for Food Recognition and Decision Support in Smart Ovens” (Electronics, 15(16):3533), this study developed an open-set metric learning framework using a modified MobileNetV2 architecture for food recognition and unknown object rejection in smart kitchen appliances. By replacing the traditional softmax layer with normalized feature embeddings evaluated via cosine similarity against stored class centroids, their system rejects unknown food items below a calibrated similarity threshold. While Pehlivan and Kara utilize a MobileNetV2 backbone, their application domain focuses on one-shot recognition of prepared food dishes within automated domestic cooking cavities. In contrast, our study evaluates whole biological produce across multi-source external photographic repositories under natural post-harvest distribution shifts.

Jansi and Kavitha — PEOS-Net (2026) [23]: Titled “PEOS-Net: A Prototype-Energy Open-Set Network for Robust Tomato Leaf Disease Recognition Beyond the Closed-World Assumption” (Frontiers in Artificial Intelligence, Volume 9, Article 1938398; published October 30, 2026), this peer-reviewed study introduced a prototype-energy open-set network for crop leaf disease recognition using a frozen EfficientNet-B0 backbone. The architecture couples a prototype-based classification branch with an energy-based OOD scoring branch, evaluated on an eight-class tomato subset of the PlantVillage repository, where two tomato disease classes were held out as unknown categories. The study incorporates held-out calibration and test partitions, multiple random seeds, and statistical evaluations. An important limitation of the study is that the unknown classes were drawn from the identical PlantVillage repository, crop species, and imaging environment as the known training classes. Consequently, PEOS-Net represents a within-dataset class-holdout evaluation and does not establish external field-condition robustness. Furthermore, because foliar pathology

involves localized planar lesions on leaves rather than whole-fruit volumetric senescence, its task setting differs fundamentally from post-harvest produce grading.

Yan et al. — PAG-MSBNet (2026) [24]: Titled "PAG-MSBNet: Prototype-aware gated multi-scale branch network for open-set crop disease recognition" (Ecological Informatics, Volume 95, Article 103703, 2026), this study formulated a multi-scale spatial-channel branch network incorporating prototype-aware gating mechanisms to classify known crop diseases while rejecting novel pathologies across public plant disease benchmarks, including PlantVillage and Paddy Doctor. Yan et al. focus on custom architectural engineering for open-set crop leaf diagnostics. In contrast, our study evaluates a standard, frozen commercial vision backbone (MobileNetV2) across external photographic repositories of harvested fruits, quantifying empirical failure modes under source-stratified covariate, species, and post-harvest treatment shifts without modifying the underlying model architecture.

2.5. Critical Literature Comparison and Research Gap

To synthesize the methodological landscape and clarify the precise scientific positioning of this study, Table 1 provides a structured comparison of recent representative studies across agricultural computer vision, domain adaptation, and open-set recognition.

2.5.1. Critical Synthesis and Specific Research Gap

The studies reviewed address complementary problems in fruit-quality classification, domain adaptation, chemical-sensor-assisted adulteration detection, and open-set recognition. The present study examines a particular combination of evaluation dimensions: established post-hoc OOD scores applied to a common frozen fruit-quality classifier, evaluated across external known-class, unseen-species, treatment-labelled, and combined-shift cohorts, with source-matched comparisons and fixed validation-calibrated operational thresholds.

Specifically, the present study investigates four evaluation dimensions:

1. Supervised Closed-Set Classification vs. External OOD Evaluation: Existing fruit-freshness studies predominantly report supervised classification performance under random train/test partitions within single curated datasets [4–6, 10]. Such evaluations do not assess how classifiers behave when exposed to external repositories or out-of-distribution instances. Our study addresses this gap by assessing a frozen classifier on external multi-repository cohorts and evaluating post-hoc OOD detection.
2. Source-Matched Cohort Comparisons to Mitigate Cross-Repository Confounding: In open-set evaluations where out-of-distribution cohorts are drawn from an external repository distinct from the training set, differences in camera hardware, ambient illumination, and background context present a substantial methodological risk of being confounded with true semantic divergence [15]. Source-matched comparisons are conducted within the same repository to reduce repository-level confounding, though potential within-repository capture variations may remain.
3. Empirical Evaluation of Treatment-Labelled Images: While multimodal systems (such as SensorNet [20]) incorporate specialized chemical gas sensors to identify adulteration, our study evaluates whether standard RGB imagery contains sufficient optical separation signal for post-hoc OOD scoring methods applied to a standard frozen classifier to distinguish treatment-labelled images (documented under a dataset-reported 10% formalin immersion protocol [12]) from natural produce, without claiming independent chemical detection.
4. Distinguishing Retrospective Ranking from Operational Threshold Calibration: Many OOD detection studies prioritize retrospective metrics: AUROC assesses ranking discrimination across all possible decision thresholds without selecting an operating point, whereas FPR@95 evaluates the false positive rate at a retrospective threshold chosen post-hoc to achieve 95% true positive rate on the test cohort [9, 15, 18]. While suitable for relative algorithm comparison, retrospective metrics do not reflect operational deployment, where a discrete acceptance threshold must be pre-calibrated on validation data and held fixed during external evaluation. Measuring operational Unknown False Acceptance Rates

Table 1. Structured literature comparison of representative agricultural computer vision, domain adaptation, and open-set recognition studies alongside the present investigation. This table reflects a literature-based comparison of reported study characteristics; it does not represent a direct quantitative head-to-head evaluation across incompatible datasets and evaluation protocols.

Study & Venue	Application Domain	Input Data & Modality	Core Methodology	Distinction from Present Study
Sultana et al. (2022) [11] Data in Brief	Fruit quality dataset curation	RGB images (16 fresh/rotten classes across 8 fruit types)	Photographic dataset collection from retail markets and fields across 16 fresh/rotten classes (8 fruit types)	Dataset descriptor publication. Evaluated as an external benchmark cohort in our study ($N = 3,193$ audited image records).
Bijoy et al. (2025) [12] Data in Brief	Fruit quality and adulteration dataset	RGB images (Fresh, rotten, formalin-mixed across 5 fruits)	Photographic dataset collection across 3 mobile smartphone cameras; augmentations	Dataset descriptor publication. Evaluated as our primary multi-shift benchmark repository ($N = 10,045$ audited image records).
Abayomi-Alli et al. (2024) [10] Multimed. Tools Appl.	Fruit freshness dataset curation	RGB images (11 fruit types; Fresh, Mild, Rotten)	Evaluates 5 baseline deep architectures (ShuffleNet, SqueezeNet, EfficientNet, ResNet18, MobileNetV2)	Multi-fruit freshness benchmark. Evaluated under closed-set classification without post-hoc OOD detection or external domain shift analysis.
Wang et al. (2023) [21] Complex & Intell. Syst.	Cross-domain fruit classification	RGB images (Grape 4-domain set; 11-fruit 2-domain set)	HAM-MobileNet (MobileNetV3 + hybrid attention) + Unsupervised Domain Adaptation (LMMD + CORAL)	Focuses on active model retraining and domain alignment. Present study evaluates a frozen classifier and post-hoc OOD detectors.
Sattar et al. (2024) [20] Intell. Syst. with Appl.	Fruit chemical adulteration detection	Multimodal: RGB images + electronic chemical gas sensors	SensorNet: hybrid CNN visual feature extraction coupled with chemical sensor array	Multimodal hardware framework integrating chemical gas sensors. Present study investigates optical RGB vision limits under treatment labels.
Pehlivan & Kara (2026) [22] Electronics	Food recognition in smart kitchen appliances	RGB images (Prepared food dataset)	Modified MobileNetV2 with open-set metric learning (cosine similarity thresholding)	Evaluates prepared food dishes in domestic cooking cavities. Present study evaluates whole fruit spoilage across external photographic repositories.
Jansi & Kavitha (2026) [23] Frontiers in Artif. Intell.	Open-set crop disease recognition	RGB images (PlantVillage tomato leaf dataset)	PEOS-Net: Prototype learning + energy-based OSR on frozen EfficientNet-B0; held-out classes	Within-dataset class-holdout evaluation on foliar lesions. Present study evaluates whole post-harvest fruit senescence across external repositories.
Yan et al. (2026) [24] Ecological Informatics	Open-set crop disease recognition	RGB images (Paddy Doctor, PlantVillage crop leaves)	PAG-MSBNet: Prototype-aware gating + multi-scale spatial-channel branch network	Architectural engineering for crop leaf pathology. Present study evaluates established post-hoc OOD scorers on a frozen standard backbone.
Present study (V4.1 evaluation)	Diagnostic evaluation of post-harvest fruit vision	Audited RGB image census ($N = 13,465$ image records; Primary, FruitVision, Sultana)	Diagnostic benchmark: 5 post-hoc OOD detectors (MSP, Energy, ODIN, Maha, OpenMax) on frozen MobileNetV2; Dual-Gate cascade	Compares source-matched cohorts to reduce repository confounding under covariate, species, and treatment shifts; bootstrap CIs; pHash auditing.

(UFAR) alongside Known-Class Rejection Rates (KRR) under fixed validation thresholds provides complementary, deployment-context measurements of operational trade-offs, supported by 1,000-iteration

non-parametric bootstrap confidence intervals and perceptual-hash (pHash) near-duplicate sensitivity analysis across an audited 13,465-record image census.

3. Dataset and Experimental Design

This section describes the source-stratified benchmark census comprising $N = 13,465$ images across three repositories, detailed cohort definitions (Partitions A, B, C, D1, D2), exact-duplicate auditing, and perceptual hash (pHash) sensitivity analysis.

4. Methodology

This section details the frozen MobileNetV2 vision architecture, the five post-hoc OOD detection scoring functions (MSP, Energy, ODIN, Mahalanobis distance, OpenMax), the operational Dual-Gate cascade, and the validation threshold calibration protocol (τ_{95}).

5. Experimental Results and Failure Analysis

This section presents the closed-set performance evaluation across partitions, source-stratified OOD detection AUROC and FPR@95 metrics, operational Dual-Gate rejection rates (UFAR, UFAR-Fresh, KRR), and non-parametric bootstrap confidence intervals.

6. Discussion

This section discusses the implications of external domain collapse, the visual limitations of RGB imaging for detecting chemical immersion protocols, and practical design recommendations for integrating secondary multi-modal sensor streams into agricultural grading pipelines.

7. Limitations and Threats to Validity

This section explicitly outlines key study limitations: reliance on image records rather than verified physical specimens, dependency on dataset-reported labels without independent biochemical assays, and lack of live in-transit cold-chain deployment testing.

8. Conclusion

This section summarizes the principal empirical findings, emphasizes the necessity of source-stratified evaluation protocols for agricultural vision benchmarks, and highlights prospective research directions in multi-sensor fusion for post-harvest quality assurance.

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